Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026 **Topic 19**

Sub-Gaussian Processes and Dudley's Entropy Integral Bound (Section 5.3)

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1 Recall: Dudley's Entropy Integral Bound for Finite Sets

Theorem 1.1 (Finite-set chaining bound)**.** *Let $\{X_\theta \mid \theta \in T\}$ be a sub-Gaussian process with respect to the metric ρ_X, and define*

$$D := \sup_{\theta,\widetilde{\theta}\in T} \rho_X(\theta,\widetilde{\theta}) < \infty.$$

For $0 \le s \le D$, recall the truncated Dudley entropy integral

$$J(s;D) := \int_s^D \sqrt{\log N(u;T,\rho_X)}\, du,$$

where $N(u;T,\rho_X)$ denotes the covering number of T at radius u.
Let $U \subseteq T$ satisfy $0 < |U| < \infty$, and define

$$L := \min\left\{i \ge 1 : N\left(D\,2^{-i};T,\rho_X\right) \ge |U|\right\}.$$

Then

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in U} \left|X_\theta - X_{\widetilde{\theta}}\right|\right] \le 8\sqrt{2}\int_{D\,2^{-(L+1)}}^{D/2} \sqrt{\log N\left(u;T,\rho_X\right)}\, du = 8\sqrt{2}\, J\left(D\,2^{-(L+1)};D/2\right).$$

Here, $N(u;T,\rho_X)$ denotes the covering number of T with respect to ρ_X at radius u.

Proof. The proof was established in the preceding lecture and is not repeated here.

2 Expected Suprema over Arbitrary Index Sets

Recall that the expected supremum is defined by

$$\mathbb{E}\sup_{t\in T} X_t := \sup\left\{\mathbb{E}\left[\sup_{t\in U} X_t\right] \,\middle|\, U \subseteq T,\ 0 < |U| < \infty\right\}.$$

Lemma 2.1 (Expected supremum over finite subsets)**.** *Let T be a nonempty index set, and let $\{X_t : t \in T\}$ be a family of integrable real-valued random variables on a common probability space. Define*

$$\mathbb{E}\left[\sup_{s,t\in T}(X_s - X_t)\right] := \sup_{\substack{F\subseteq T\times T\\ 0<|F|<\infty}} \mathbb{E}\left[\max_{(s,t)\in F}(X_s - X_t)\right].$$

Then

$$\mathbb{E}\left[\sup_{s,t\in T}(X_s - X_t)\right] = \sup_{\substack{U\subseteq T\\ 0<|U|<\infty}} \mathbb{E}\left[\max_{s,t\in U}(X_s - X_t)\right].$$

The equality is understood in the extended-real sense, so the common value may be $+\infty$.

Proof. This is Homework Problem.

3 Dudley's Entropy Integral Bound

Theorem 3.1 (Dudley's entropy integral bound)**.** *Let $\{X_\theta \mid \theta \in T\}$ be a sub-Gaussian process with respect to ρ_X, and suppose that*

$$D := \sup_{\theta,\widetilde{\theta}\in T} \rho_X(\theta,\widetilde{\theta}) < \infty.$$

Then

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left(X_\theta - X_{\widetilde{\theta}}\right)\right] \le 8\sqrt{2}J(0;D).$$

Proof. By Lemma 2.1 and Theorem 1.1,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left(X_\theta - X_{\widetilde{\theta}}\right)\right] &= \sup_{\substack{U\subseteq T\\ 0<|U|<\infty}} \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in U}\left(X_\theta - X_{\widetilde{\theta}}\right)\right]\\ &\le \sup_{\substack{U\subseteq T\\ 0<|U|<\infty}} 8\sqrt{2}J(0;D)\\ &= 8\sqrt{2}J(0;D).\end{aligned}$$

□

Remark 3.2. For every realization of the process,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right] &= \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T} max\{X_\theta - X_{\widetilde{\theta}},\ X_{\widetilde{\theta}} - X_\theta\}\right]\\ &= \mathbb{E}\left[\max\{\sup_{\theta,\widetilde{\theta}\in T}(X_\theta - X_{\widetilde{\theta}}),\ \sup_{\theta,\widetilde{\theta}\in T}(X_{\widetilde{\theta}} - X_\theta)\}\right]\\ &= \mathbb{E}\left[\max\{\sup_{\theta,\widetilde{\theta}\in T}(X_\theta - X_{\widetilde{\theta}}),\ \sup_{\widetilde{\theta},\theta\in T}(X_{\widetilde{\theta}} - X_\theta)\}\right]\\ &= \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left(X_\theta - X_{\widetilde{\theta}}\right)\right]\end{aligned}$$

Indeed, interchanging θ and $\widetilde{\theta}$ changes the sign of the increment. Consequently, the conclusion of Theorem 3.1 can equivalently be written as

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right] \le 8\sqrt{2}J(0;D).$$

Remark 3.3. Fix any $\theta_0 \in T$. By the triangle inequality,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta\in T}|X_\theta|\right] &= \mathbb{E}\left[\sup_{\theta\in T}|X_\theta - X_{\theta_0} + X_{\theta_0}|\right]\\ &\le \mathbb{E}\left[\sup_{\theta\in T}|X_\theta - X_{\theta_0}|\right] + \mathbb{E}\left[|X_{\theta_0}|\right]\\ &\le \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right] + \mathbb{E}\left[|X_{\theta_0}|\right].\end{aligned}$$

Remark 3.4. If there exists $\theta_0 \in T$ such that $X_{\theta_0} = 0$ almost surely, then

$$\mathbb{E}\left[\sup_{\theta\in T}|X_\theta|\right] \le \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right]$$

4 Application to Rademacher Complexity

Example 4.1 (The Rademacher process)**.** Fix $x_1, \dots, x_n \in \mathcal{X}$, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher random variables. For $f \in \mathcal{F}$, define

$$Z_f := \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varepsilon_i f(x_i).$$

The process $\{Z_f \mid f \in \mathcal{F}\}$ is sub-Gaussian with respect to $\|f - g\|_{P_n}$

$$\rho(f, g) := \|f - g\|_{P_n} := \left(\frac{1}{n} \sum_{i=1}^{n} (f(x_i) - g(x_i))^2 \right)^{1/2}.$$

The conditional empirical Rademacher complexity is

$$\begin{aligned} \mathcal{R}\left(\frac{\mathcal{F}(x_1^n)}{n} \right) &:= \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(x_i) \right| \right] \\ &= \frac{1}{\sqrt{n}} \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} |Z_f| \right]. \end{aligned}$$

Suppose there exists $f_0 \in \mathcal{F}$ such that

$$f_0(x_1) = f_0(x_2) = \cdots = f_0(x_n) = 0.$$

Then $Z_{f_0} = 0$ almost surely.
By Theorem 3.1 and Remark 3.4,

$$\mathcal{R}\left(\frac{\mathcal{F}(x_1^n)}{n} \right) \leq \frac{8\sqrt{2}}{\sqrt{n}} \int_0^D \sqrt{\log N(u; \mathcal{F}, \|\cdot\|_{P_n})}\, du,$$

where

$$D := \sup_{f, g \in \mathcal{F}} \|f - g\|_{P_n}.$$

Example 4.2. Consider the function class

$$\mathcal{F} := \left\{ \mathbf{1}_{(-\infty, t]}(\,\cdot\,) : t \in \mathbb{R} \right\}.$$

The identically zero function is not an element of $\mathcal{F}$. However, for a fixed sample $x_1^n = (x_1, \dots, x_n)$, choose

$$t_0 < \min_{1 \leq i \leq n} \{x_i\} \qquad \text{and set} \qquad f_0 := \mathbf{1}_{(-\infty, t_0]}(\,\cdot\,).$$

Then $f_0 \in \mathcal{F}$ and

$$f_0(x_1) = f_0(x_2) = \cdots = f_0(x_n) = 0.$$

Suggested reading. See Sections 5.4 and 5.5 of [1] for further discussion of Gaussian processes.

References

[1] M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.